Functional Dependencies and Normalization

LumoraAI Presentation Demo • 03 DBMS

Functional dependency

• A functional dependency X → Y states that if two tuples agree on attributes X, they must also agree on attributes Y.

Normal forms

• First Normal Form requires atomic attribute values rather than repeating groups. Second Normal Form removes partial dependency of non-key attributes on part of a composite candidate key. Third Normal Form targets transitive dependencies from keys through non-key attributes.

Why normalize

• Normalization can reduce update, insertion and deletion anomalies by separating facts that should be represented independently. Over-normalization can increase the number of joins, so real designs balance integrity and query needs.

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